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Optimizing pollutant exposure, energy consumption, and thermal comfort in a house via deep reinforcement learning control

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ABSTRACT

Efficient indoor environment management is challenging due to the complex interdependent nature of pollutant levels, energy use, and thermal comfort. While data-driven and physics-based algorithms have been explored to optimize these conflicting objectives, their real-world integration is constrained by modeling, computational, and learning challenges. Reinforcement learning (RL) has gained attention as a potential substitute in such scenarios. This work developed a reward-based deep RL agent for optimizing pollutant exposure, energy consumption, and thermal comfort by controlling ventilation rates and set temperature of the heating, ventilation, and air conditioning (HVAC) unit. Two RL agents were developed: Agent 1 controlled only the ventilation rate, and Agent 2 controlled both the ventilation rate and the HVAC set temperature. A reward function with weighted combinations of energy (W_1), pollutant exposure (W_2), and thermal discomfort (W_3) enabled specific control strategies with and without an HVAC filter. RL agents' reliability under varying pollutant emission scenarios was evaluated against a physics-based dynamic optimization (DynOpt) strategy. Agent 1 achieved ~26 %–~133 % higher normalized exposure reduction (NER) than DynOpt at a W_1/W_2 ratio of 1/3. On the other hand, Agent 2 ($W_1/W_2 = 1/10$, and $W_3 = 1$ or 10) dynamically adjusted the set temperature between 25 °C and 27°C, achieving NER values ~17 %–~373 % higher than Agent 1 across all reward combinations. The proposed algorithm can be deployed in the field through low-cost microcontrollers and monitors, presenting a potentially deployable solution for healthy indoor environments with minimal environmental impacts.

1. Introduction

Buildings impact occupants' health, well-being, and productivity since people spend almost 90 % of their time inside buildings [1,2]. Though indoor environments act as a shield against outdoor elements and pollutants, the exposure to pollutants could be considerably higher indoors than outdoors [3–5]. Simultaneously, indoor spaces may lead to a higher risk of infectious disease transmission and may trap airborne pathogens, leading to prolonged exposure [6–10]. An effective way to reduce pollutant exposure and prevent disease transmission in buildings is to increase the indoor-outdoor air exchange rate (AER). However, increasing AER is

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Abbreviations

ACH	Air Changes per Hour
AER	Air Exchange Rate
AHU	Air Handling Unit
DASS	Dedicated Air Supply System
DI	Discomfort Index
DOPEEC	Deep reinforcement learning agent for Optimizing Personal Exposure, Energy consumption, and thermal Comfort
DQN	Deep Q-Network
DRL:	Deep Reinforcement Learning
DynOpt	Dynamic Optimization
HE	High Emission
HVAC	Heating, Ventilation, and Air Conditioning
IAQ	Indoor Air Quality
LE	Low Emission
MPC	Model Predictive Control
NER	Normalized Exposure Reduction
NO_X	Nitrogen Oxide
PM	Particulate Matter
RL:	Reinforcement Learning

only feasible when ambient pollutant concentration is low; otherwise, it further exacerbates indoor air quality (IAQ). Studies have reported that outdoor-origin pollutants can dominate or contribute equally to exposure in indoor spaces [11,12].

The increased AER for IAQ improvement can also compromise occupants' thermal comfort if the air brought inside is not conditioned, which requires additional energy consumption for the heating, ventilation, and air conditioning (HVAC) systems. This additional energy demand could be unsustainable, considering that the HVAC systems already account for ~50 % of buildings' energy consumption [13–15]. Therefore, a trade-off exists between pollutant exposure, energy consumption, and thermal comfort, which necessitates optimization of indoor-outdoor AER and operation of HVAC systems. Recent studies have proposed strategies that exploit physics-based domain knowledge, such as model predictive control (MPC) [16,17] and dynamic optimization [18–20], to control the operation of HVAC and ventilation systems to optimize the energy-exposure trade-off. For example, Mishra et al. [18] developed a real-time dynamic optimization strategy for optimizing indoor particulate matter (PM) and energy consumption. The study demonstrated a positive correlation between energy consumption and exposure reduction, i.e., an increase in energy consumption (energy penalty) accompanied exposure reduction compared to the base scenario. The same study also achieved a reduced energy penalty or even savings, with minimal or no thermal discomfort to the occupants, when the indoor set temperature was increased by 1 °C. Ganesh et al. [20] worked on the optimization of energy consumption of a dedicated ventilation unit while ensuring that pollutant concentration remains below desirable limits, though the study did not consider thermal comfort in the optimization problem. This study demonstrated that the optimization reduced peak pollutant concentration by ~31 % and energy consumption by ~18 %. Sha et al. [21] developed a data-driven model predictive control to optimize thermal comfort, energy consumption of HVAC, and indoor CO_2 concentration. The proposed strategy reduced peak CO_2 concentration by ~23 %, while also reducing the energy consumption by ~10 %.

However, optimization strategies such as dynamic optimization and MPC require physics-based models for indoor pollutant dynamics and HVAC operation. The indoor pollutant and HVAC operation models further require detailed house characteristics such as house dimensions, thermal permeability, and pollutant deposition and infiltration rates. These input parameters either need to be experimentally obtained or modeled for each site, therefore making it infeasible to deploy physics-based models at a large scale. Thus, designing and developing algorithms for individual households for the wide-scale field deployment of fully physics-driven agents is challenging. Therefore, researchers have used data-driven algorithms to predict HVAC energy usage, air conditioning load, indoor air quality, and thermal comfort [22–26]. Despite their advantages, the utility of traditional data-driven models is limited in predicting control actions in highly dynamic and continuously evolving systems such as indoor environments. Moreover, incorporating physical constraints such as AER bounds, actuator ranges, and power consumption is challenging.

Contrarily, deep reinforcement learning (DRL) agents have shown the potential to control a range of indoor environments [27–36], and they can overcome many challenges associated with purely physics-driven and data-driven strategies. Learning the optimal policies directly through interacting with the environment develops the agent's understanding of their action influence on the system dynamics [37]. The exploration-exploitation-feedback technique ensures the adaptability of RL agents to changing system dynamics [38]. These capabilities make DRL agents suitable for control tasks in indoor environments. Guo et al. [28] showed that the DRL controller reduced energy consumption by ~21 % compared to a rule-based controller with improved thermal comfort and reduced indoor CO_2 concentration when co-optimizing thermal comfort, energy consumption, and IAQ. Yao et al. [39] proposed an HVAC control strategy based on DRL and fuzzy reasoning to optimize HVAC supply air temperature, flow rate, and velocity. This study achieved ~6 % higher energy cost savings and ~15 % better thermal comfort compared to RL methods that only optimize HVAC supply air temperature. Similar studies in the field have optimized the operation of the HVAC system to reduce energy consumption and personal exposure [30,40]. However, the control action in most studies is limited to indoor-outdoor AER, and only a few studies have dynamically varied HVAC supply air temperature or indoor set temperature [39,41]. A dynamic indoor set temperature can reduce the energy consumption required for air conditioning while ensuring the thermal comfort of the occupants [18,41]. Thus, the dynamic control of indoor-outdoor AER and HVAC set temperature allows for the operation of an energy-efficient and healthy indoor environment, minimizing personal exposure to airborne pollutants and pathogens, and mitigating airborne transmission risk. However, limited work has been done on developing controls for indoor-outdoor AER and HVAC set temperature for concurrent optimization of PM exposure, energy consumption, and thermal comfort.

This study develops a deep reinforcement learning (DRL) agent for indoor environment control of a house [42] with indoor-outdoor AER and HVAC set temperature as control actions. The agent is called a Deep reinforcement learning agent for Optimizing Personal Exposure, Energy, and Comfort (DOPEEC). Real-time indoor and outdoor pollutant concentration, energy consumption, temperature, and RH collected through low-cost monitors are fed to DOPEEC to control the operation of the dedicated air supply system (DASS) and the indoor set temperature of the air conditioning unit. Longitudinal particulate matter (PM) emission rate data from the house are used to train DOPEEC, and data augmentation techniques are incorporated to increase the data density for training. Firstly, the DOPEEC controls the indoor-outdoor AERs via DASS operation to minimize the air conditioning load while reducing pollutant exposure and maintaining thermal comfort (Agent 1). Subsequently, a comparative assessment of the DOPEEC with a physics-based dynamic optimization (hereafter referred to as DynOpt) strategy proposed by Mishra et al. [18] is performed. Secondly, the DOPEEC is modified and trained to change the set temperature of the air conditioning unit along with the indoor-outdoor AER (Agent 2). Similar comparative assessments as Agent 1 are performed to evaluate the performance of Agent 2. Specifically, the performance of the DOPEEC is assessed based on (i) cumulative exposure reduction, (ii) total energy consumption, and (iii) normalized exposure reduction (NER: percentage exposure reduction normalized by percentage increase in energy consumption). Ultimately, this work advocates for adopting and integrating control strategies in buildings to optimize various elements governing the indoor environment quality and energy consumption for a sustainable and comfortable future.

2. Methodology

The DOPEEC outputs the indoor-outdoor AER (Agent 1) and indoor-outdoor AER and set temperature (Agent 2) using the inputs, including ambient and indoor temperature, relative humidity (RH), and indoor PM concentration. The DOPEEC is integrated with a digital twin of the house created using the experimental data obtained from a field study (discussed in Section S1 of the Supplementary Information (SI)), in OpenAI Gym, a Python API for creating training environments in reinforcement learning. More details about OpenAI Gym can be found at [43]. The following sections present a brief discussion of the house and emission parameters, the development of the DRL agent, dataset availability, and comparative assessment with a dynamic optimization strategy.

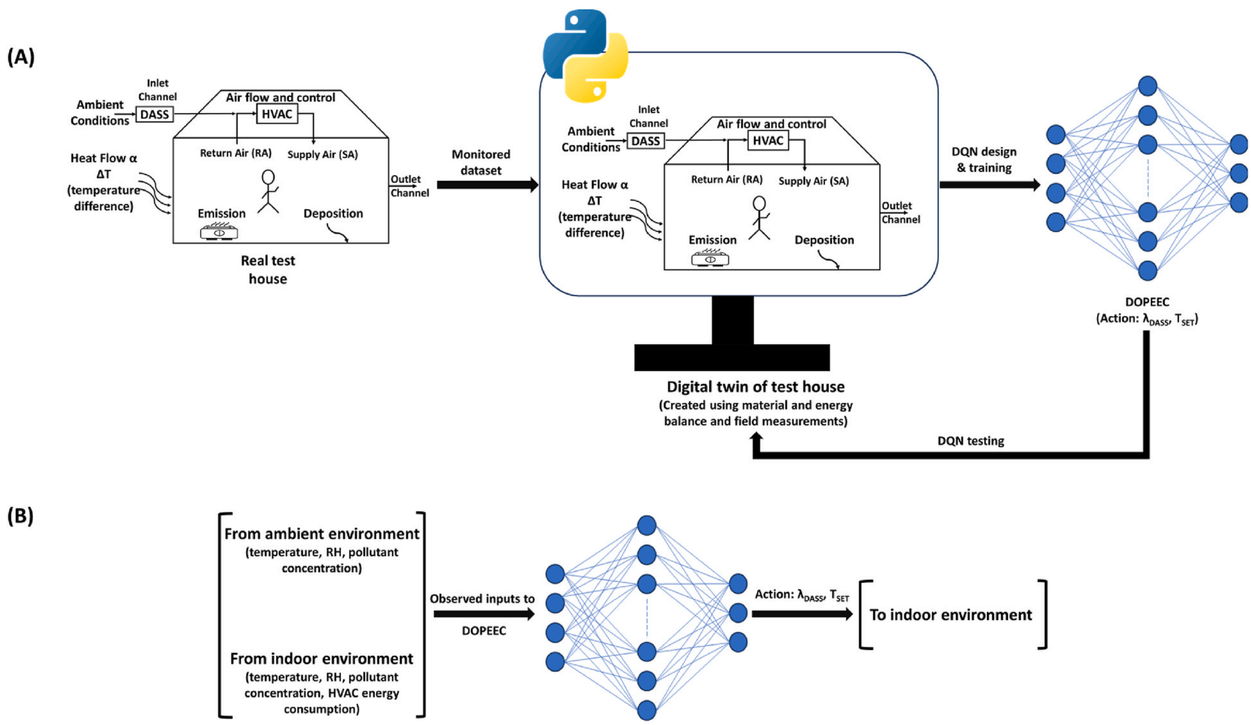

Fig. 1. (A) A framework for creating the digital twin of the house and controlling the dedicated air supply system (DASS) and indoor set temperature for training and testing the deep reinforcement learning (DRL) agent, and (B) input/output parameters of the DRL agent. λ_{DASS} is the indoor-outdoor air exchange rate, and T_{SET} is the indoor set temperature.

2.1. Deep reinforcement learning approach and dynamic optimization

2.1.1. Digital twin of the house, general framework, and data curation for DOPEEC training

A digital twin of the house needed for offline training of the DOPEEC is constructed using the measured and derived parameters from a field investigation [42] fed to the particle dynamics and energy balance models. The nine high and low pollutant emission scenarios have been split into a 2:1 ratio for training and testing DOPEEC, i.e., six days for training and three for testing. The parameters monitored during the six experimental days of the field study have been augmented to increase the data density for training. A framework for developing DOPEEC to control the DASS operation and set temperature is demonstrated in Fig. 1A. A brief discussion of monitored parameters during the field study, data augmentation techniques, and data curation has been included in Section S2 of the SI. A detailed discussion on creating the digital twin using field measurements and derived parameters is also present in Sections S2, S3, and S4 of the SI.

Briefly, the three-bedroom test house has a floor area of ~111 m^2 and a volume of ~250 m^3. All measurements concerning this work were performed with all external doors and windows closed while DASS maintained an outdoor-indoor AER of $0.5 \pm 0.1\ h^{-1}$. The air handling unit (AHU) of the HVAC system operated continuously without a filter. The continuous AHU flow rate of 2000 $m^3\ h^{-1}$, equivalent to 8 air changes per hour (ACH), minimized spatial gradients. During the field study, two kinds of experiments were performed: low-emission (LE) and high-emission (HE). LE activities were further bifurcated into two different categories: (1) LE1, where a real-life day was replicated by cooking three meals (breakfast, lunch, and dinner) and floor mopping, and (2) LE2, where one dish was cooked multiple times throughout the day at regular intervals [42]. HE activities included preparing a typical American Thanksgiving meal for about 14 people. Particulate matter (PM) source strengths, infiltration rates, and deposition rates estimated using the measurements by Patel et al. [44,45] for nine different experimental days (four LE1s, two HEs, and three LE2s) have been used as input for this study. A comprehensive description of the field study design has been presented by Farmer et al. [42].

The indoor pollutant concentration measured during the nine experimental days of the field study was used to create the pollutant balance model, while the energy balance was utilized to replicate the thermal environment inside the house. The pollutant balance and thermal model match well with the measurements in the house. The digital twin has been created to train the model, and no physical laws have been fed to the DOPEEC. The simulated indoor measurements (temperature, RH, PM concentration, and energy consumption) and measured outdoor parameters (temperature, RH, and PM concentration) from the digital twin act as input to the DOPEEC during the training and prediction phases. Fig. 1B demonstrates the input/output parameters of the DOPPEC for training and testing and corresponding output/control actions.

2.1.2. Benchmark and dynamic optimization (DynOpt)

The digital twin of the house described in the above sub-section generates temporal indoor PM profiles by simulating the emission sources and operation of DASS and HVAC systems corresponding to the house operating conditions. The simulated results agree well with the measured data (Fig. S9–S11). The recreated PM concentration profile, total exposure, and total energy consumption are used as benchmarks for all comparative analyses in this work. The benchmark data and its comparison with field measurements are discussed in Section S5 of the SI.

The DynOpt strategy aims to control DASS and HVAC operation to optimize the energy-exposure trade-off and ensure thermal comfort. The objective function is expressed mathematically in Eq. (1). The first component represents energy consumption, and the second component corresponds to indoor PM exposure. W_1 and W_2 are user-assigned weightages for the energy and exposure components of the objective function, and $C_{\max}$ is the defined threshold for indoor PM concentration.

$$\textit{minimize}$$

$$obj = W_1 E + W_2(\max(0, C - C_{max}))^a \tag{1}$$

$$\text{subjected to } \lambda_{min} \leq \lambda_{DASS} \leq \lambda_{max}$$

Where λ_{min} (0.5 h^{-1}) and $\lambda_{\max}$ (10 h^{-1}) are the lower and upper values of the indoor-outdoor AER, respectively. Mishra et al. [18] discussed the selection of W_1/W_2, $C_{\max}$, and a-values in detail, and Section S6 in the SI briefly summarizes the same.

2.1.3. Deep reinforcement learning agent

The developed DOPEEC in this work is based on Deep Q-Network (DQN) learning, which is trained and tuned to minimize the same objective function as dynamic optimization. The DOPEEC interacts continuously with the environment and learns an optimal decision-making policy for reducing pollutant exposure and optimizing associated energy penalties to ensure thermal comfort. The agent has three components: the state (s), the action (a), and the reward (r). At any time, the agent observes the state s_t, takes action, a_t, according to the policy based on a value function, and the state changes to s_{t+1}. The process of selecting an action by the agent is termed policy, often denoted by $\pi_t\ (s_t|a_t)$, and explains the decision of an action at any given state. The returns of taking an action at any state are estimated through the value function ($Q_\pi(s, a)$), shown in Eq. (2) [33].

$$Q_\pi(s,a) = E\left[r + \gamma \max_{a'} Q_\pi(s,a)\right], \forall s \in S, \forall a \in A \tag{2}$$

where $E[]$ is the expectation of the expression inside the bracket estimated for a given policy $\pi_t\ (s_t|a_t)$, S is the possible set of state space, and A is the feasible set of actions. The agent receives a reward r_t depending on the evaluation of the action a_t. The policy is updated to maximize the total reward (G_t), accounting for the future rewards by a discount factor γ as expressed in Eq. (3). The agent collects experiences from the environment and improves its policy through self-learning based on these experiences, thus making it suitable for dynamic systems [31,32].

$$G_t = r_t + \gamma\, r_{t+1} + \gamma^2\, r_{t+2} \; + \ldots = \sum_{k=t}^{\infty} \gamma^{k-t}\, r_t \tag{3}$$

Fig. 2 demonstrates a schematic for the training and performance evaluation of the agent. The DQN consists of two dense neural networks. The behavior network with weights, w_b, takes the decision and interacts with the environment, and the target network with weights, w_t, helps update the behavior network [31,32,36]. The DQN utilizes a replay buffer that stores and replays experiences gathered from agent interactions with the environment. A random batch sampling from the replay buffer avoids overfitting, promotes learning from past mistakes, and helps eliminate the correlation within the input data in each batch. At each step, the state (s_t) is fed as an input to the behavior network, and the agent decides an action (a_t) depending on the output of the behavior network (Q_b (s_t, a_t*)) and the ε-greedy strategy, where a_t* represents all possible actions. The ε-greedy strategy refers to selecting an action with the maximum value of Q_b (s_t, a_t*) with a probability of ε and randomly with a probability of 1-ε. The replay buffer, with a memory capacity of N, stores the current state (s_t), the current action (a_t), the received reward (r_t), and the next state (s_{t+1}). The behavior network is updated through minimizing the loss function through a random batch sampling (s_k, a_k, r_k, s_{k+1}) from the replay buffer.

The input to the behavior network is (s_k, a_k), and the output is Q_b (s_k, a_k), while for the target network, the input and output are (r_k, s_{k+1}) and maximum Q_t (s_{k+1}, a_{k+1}*). The loss function, L_k, is then estimated, as shown in Eq. (4) [31,32], and the weights of the behavior network are updated based on the estimated loss. After that, w_t (weights of the target network) is updated after every m timestep, which means replacing it with w_b (weights of the behavior network). The update frequency, m, of the target network is a trainable parameter, and its selection has been discussed later.

$$L_k = \left(r_k + \gamma \,\max Q_t\left(s_{k+1}, a^*_{k+1}\right) - Q_b(s_k, a_k)\right)^2 \tag{4}$$

The proposed DQN is applied to train two agents: (1) Agent 1, which controls the indoor-outdoor AER, and (2) Agent 2, which controls the indoor-outdoor AER and the set temperature. Both agents aim to reduce $PM_{2.5}$ exposure and ensure thermal comfort while optimizing energy consumption. For Agent 1, two cases have been considered: (i) DASS control without an HVAC filter and (ii) DASS control with an HVAC filter. Different indoor environment parameters are represented in the state s_t, as shown in Eq. (5).

$$s_t = [T_{out}(t), RH_{out}(t), C_{in}(t), \lambda_{DASS}(t), T_{in}(t), RH_{in}(t), P_{in}(t)] \tag{5}$$

where C is $PM_{2.5}$ concentration, T is air temperature, RH is relative humidity, λ_{DASS} is air changes per hour, and P is the total power consumption of HVAC and DASS, the subscripts *out* and *in* means outdoor and indoor, respectively. The state space includes one more element: the set temperature of the air conditioning unit (T_{set} (t)) in the case of Agent 2. The action (a_t) has a single element λ_{DASS} for

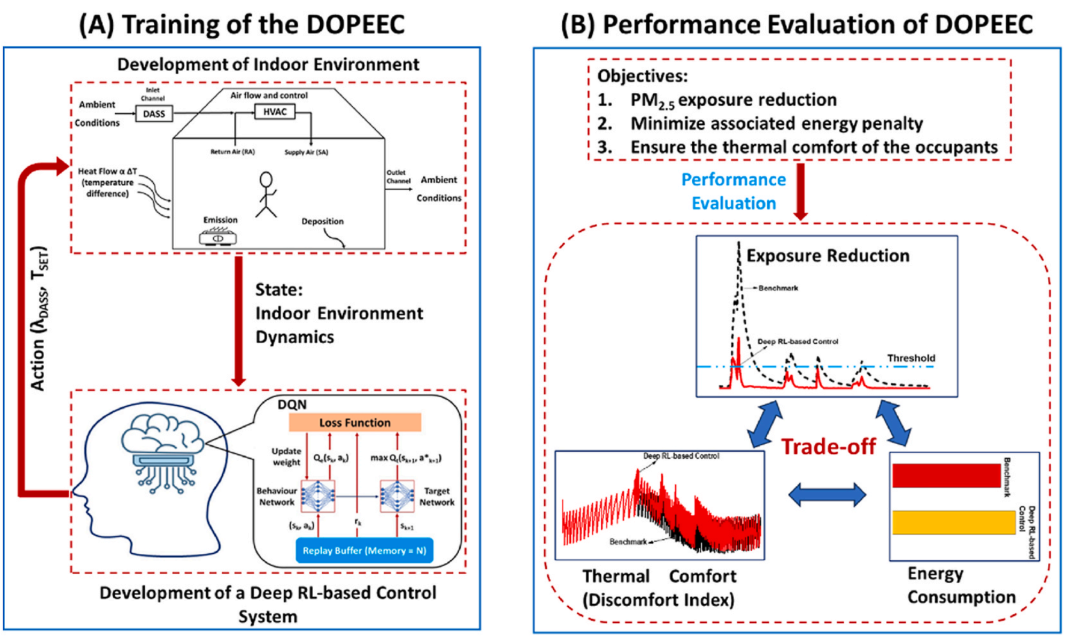

Fig. 2. Schematic of (A) training and (B) performance evaluation of the Deep reinforcement learning agent for Optimizing Personal Exposure, Energy, and Comfort (DOPEEC). λ_{DASS} is the indoor-outdoor air exchange rate, and T_{SET} is the indoor set temperature.

Agent 1 and two elements λ_{DASS} and T_{set} for Agent 2. λ_{DASS} in the action space varies between 0.5 ACH to 10 ACH with a step size of 0.5. At the same time, T_{set} can be either 25, 26, or 27 °C. Therefore, 20 and 60 actions are possible at any time for Agent 1 and Agent 2, respectively, and all possible sets of actions for both agents are shown in Eq. (6).

For Agent 1:

$$a_t = 0.5 \text{ } or \text{ } 1 \text{ } or \text{ } 1.5 \text{ } or \text{ } 2 \text{ } or \text{ } \ldots\ldots\ldots\ldots or \text{ } 10$$

For Agent 2:

$$a_t = (0.5, 25) \text{ } or \text{ } (0.5, 26) \text{ } or \text{ } (0.5, 27) \text{ } or \text{ } (1, 25) \text{ } or \text{ } (1, 26) \text{ } or \ldots\ldots (10, 27) \tag{6}$$

Agent 1 aims to maximize the reward, which is negative of the cost function defined in Eq. (1), where only energy and exposure components are present, and the set temperature is kept at 25 °C at all times to ensure occupants' thermal comfort. For Agent 2, the reward function, *R(t),* is modified to include a third term representing Thom's discomfort index (*DI*) [46,47], a measure of the thermal comfort of the occupants, and is shown in Eq. (7). The inclusion of *DI* in the reward function will lead to dynamic control of set temperature and thermal comfort in the indoor environments. More discussion on the discomfort index is present in Section S7 in the SI.

$$R(t) = -W_1 E - W_2(\max(0, C - C_{max}))^{\alpha} - W_3 DI \tag{7}$$

A brief discussion on choices of weights (W_1, W_2, W_3), $C_{\max}$, and α-values has been presented in Section S6 of the SI and discussed in detail by Mishra et al. [18]. For all simulations in this study, $C_{\max}$ is 10 μg/m^3, and α-value is 2.

3. Results and discussion

3.1. Training parameters of the DOPEEC

The training of the DOPEEC is conducted offline inside the digital twin of the house on the augmented dataset synthesized using parameters from six experimental days, as discussed in Section 2.1.1. The agent is trained for 1500 episodes on the augmented datasets, and a grid search is performed over all possible combinations of hyperparameters, such as learning rate, batch size, and number of hidden layers and nodes, shown in Table S2 of the SI, to obtain the optimal model parameters. Based on the performance of the DOPEEC, the model parameters with the best reward among all the combinations and episodes have been adopted for all further simulations in this study. More details on hyperparameter tuning and their selection are present in Section S7 of the SI. Finally, the trained agent is deployed for performance evaluation, and a comparative analysis with respect to benchmark conditions and DynOpt is performed for three experimental days (discussed in Section 2.1.1).

Table 1

Exposure, energy, and NER-value with and without HVAC filter configurations for two W_1/W_2 ratios (1/3 and 1/10) at an indoor set temperature of 25 °C (Agent 1).

Parameters				Exposure (μg h/m^3)	Energy (kWh)	NER
LE1	Benchmark			132.77	8.63	–
	$W_1/W_2 = 1/3$	Without HVAC Filter	DynOpt	83.08	9.73	2.94
			DOPEEC	104.29	8.90	6.86
		With HVAC Filter	DynOpt	27.52	10.47	3.72
			DOPEEC	30.68	9.91	5.18
	$W_1/W_2 = 1/10$	Without HVAC Filter	DynOpt	82.97	9.73	2.94
			DOPEEC	79.75	10.56	1.78
		With HVAC Filter	DynOpt	27.51	10.47	3.72
			DOPEEC	28.57	10.07	4.70
LE2	Benchmark			176.66	10.04	–
	$W_1/W_2 = 1/3$	Without HVAC Filter	DynOpt	99.85	12.78	1.59
			DOPEEC	138.95	9.17	2.46
		With HVAC Filter	DynOpt	33.82	12.62	3.15
			DOPEEC	37.83	11.41	5.76
	$W_1/W_2 = 1/10$	Without HVAC Filter	DynOpt	99.75	12.78	1.59
			DOPEEC	92.92	14.57	1.05
		With HVAC Filter	DynOpt	33.80	12.62	3.15
			DOPEEC	35.89	11.76	4.65
HE	Benchmark			860.76	9.19	–
	$W_1/W_2 = 1/3$	Without HVAC Filter	DynOpt	227.71	16.20	0.96
			DOPEEC	307.58	12.53	1.77
		With HVAC Filter	DynOpt	105.89	14.69	1.46
			DOPEEC	126.85	12.31	2.51
	$W_1/W_2 = 1/10$	Without HVAC Filter	DynOpt	227.51	16.26	0.96
			DOPEEC	224.48	17.80	0.79
		With HVAC Filter	DynOpt	105.79	14.79	1.44
			DOPEEC	112.55	13.30	1.94

3.2. Dependency of DOPEEC performance on W_1/W_2 ratio

This work studied the interplay between energy and exposure for two distinct W_1/W_2 ratios (1/3 and 1/10). Similar to dynamic optimization, for multi-objective optimization problems, DOPEEC does not minimize all objectives simultaneously, and therefore, a trade-off exists, requiring a balance of objectives. The balance is, thus, taken care of in the reward function (Eqs. (1) and (7)), where a weighted combination of all objectives is used. In the previous work by Mishra et al. [18], the sensitivity of energy-exposure interplay to weights (W_1/W_2) was performed (Fig. S13). In the case of reward-based DRL, previous studies have also defined a weighted combination of multiple conflicting objectives. However, apart from a few studies that have performed a sensitivity analysis of variations in weights assigned to different objectives [48,49], most studies have assigned pre-defined weights in the reward function [28,33,34,50,51]. Here, the performance evaluation of DOPEEC with respect to changes in the W_1/W_2 ratios is carried out based on Agent 1. For Agent 1, the indoor set temperature is kept at 25 °C at all times to ensure the thermal comfort of the occupants. Therefore, the reward/cost function for Agent 1 has two terms: energy and exposure (Eq. (1)). Thus, only the variations in W_1/W_2 ratios have been studied (W_1: weight assigned to energy component and W_2: weight assigned to exposure component) since W_3 is not applicable for Agent 1. The insights from the discussion will also remain relevant for other optimization and control strategies that utilize a weighted combination of multiple objectives.

For LE1, decreasing the W_1/W_2 ratio from 1/3 to 1/10 increased the exposure reduction by ~23 %. However, the associated energy penalty also increased by ~19 %, significantly reducing the NER from 6.86 ($W_1/W_2 = 1/3$) to 1.78 ($W_1/W_2 = 1/10$), as shown in Table 1. Similar trends in exposure reduction, energy penalty, and NER are also observed for HE and LE2. Therefore, the choice of weights in the reward/cost function is critical to ensure the optimal performance of these systems (DynOpt and DOPEEC). The performance of the DOPEEC will also change depending on the installation of the HVAC filter, as evidenced by considerable exposure reduction via the filter installation (Table 1). Therefore, for Agent 1, two cases have been formulated: (i) DASS control without an HVAC filter and (ii) DASS control with an HVAC filter. The subsequent sections discuss the performance of DOPEEC under different operating conditions and environments relative to benchmark and DynOpt.

3.3. DASS control with and without HVAC filter (agent 1)

3.3.1. DOPEEC performance: energy-exposure optimization by controlling DASS

For LE1, LE2, and HE, the indoor PM concentration profile corresponding to benchmark, DynOpt, and DOPEEC for two distinct W_1/W_2 (1/3 and 1/10) are shown in Fig. 3. During LE1, in the absence of an HVAC filter and at W_1/W_2 ratio of 1/3 (Fig. 3B), a higher exposure reduction relative to benchmark was observed in case of DynOpt (~37 %) as compared to DOPEEC (~21 %). The higher exposure reduction, however, comes at an increased energy penalty, which is reflected in a lower NER value for DynOpt (2.94) compared to DOPEEC (6.86). In addition, the peak PM concentration reduction relative to the benchmark for DOPEEC (~41 %) is also

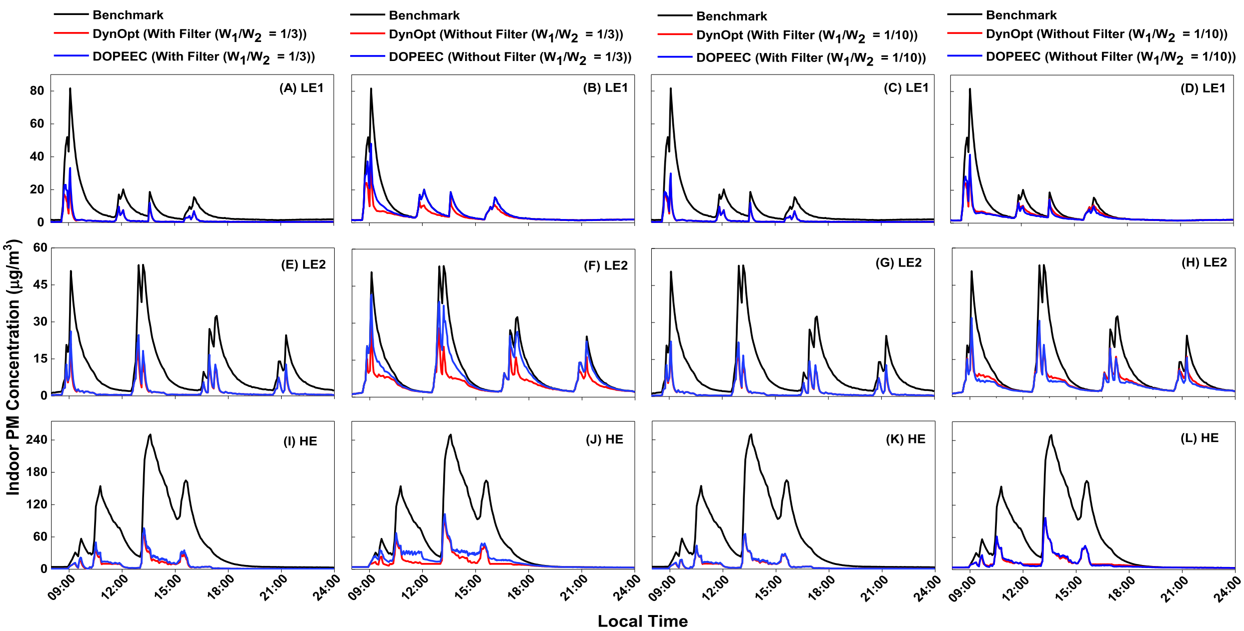

Fig. 3. Temporal indoor PM concentration profiles of benchmark (experimental; without HVAC filter), physics-based dynamic optimization (DynOpt), and reinforcement learning agent (DOPEEC) for with and without HVAC filter configurations for two W_1/W_2 ratios (1/3 and 1/10) during LE1 (A–D), LE2 (E–H), and HE (I–L). Each row corresponds to a particular experimental day, and each column corresponds to a combination of HVAC filter configuration and W_1/W_2 ratio.

lower than DynOpt (~53 %). This difference between peak concentration reduction and overlap of PM concentration profiles for DOPEEC and benchmark from 12:00 to 21:00 (Fig. 3B) could be attributed to diminishing marginal returns in exposure reduction. The effects of diminishing marginal returns have been discussed by Mishra et al. [18], where the authors discussed that as the indoor PM concentration reduces and is close to the threshold value ($C_{\max}$), further lowering the concentration comes at a higher energy penalty. Therefore, for the W_1/W_2 ratio of 1/3, DOPEEC prioritizes energy savings over exposure reduction for the duration 12:00 to 21:00 (Fig. 3B). When the W_1/W_2 ratio is decreased to 1/10 from 1/3, DOPEEC reduced the exposure by ~40 %. While the exposure reduction for DynOpt remained the same (~37 %), DynOpt demonstrated higher energy efficiency (NER: 2.94) as compared to DOPEEC (NER: 1.78), again demonstrating that a trade-off exists between energy and exposure. Since exposure reduction for DynOpt and DOPEEC are almost the same, the indoor PM concentration profiles overlap (Fig. 3D). With an HVAC filter, DOPEEC outperformed DynOpt for both W_1/W_2 ratios of 1/3 and 1/10 in terms of total energy consumption and NER while achieving similar levels of exposure reduction. Thus, the indoor PM concentration profiles overlap for DynOpt and DOPEEC (Fig. 3A and C). For the W_1/W_2 ratio of 1/3 during LE1, DOPEEC reduced exposure by ~77 %, slightly less than DynOpt (~79 %), with a NER of 5.18, demonstrating a 39 % improvement compared to DynOpt (Table 1). Previous studies have also shown that the RL agents perform better than rule-based and MPC controllers, though the performance comparison metrics are different [52]. For example, An et al. [31] developed a DRL agent to control indoor $PM_{2.5}$ while ensuring thermal comfort and discussed the window opening and closing pattern along with HVAC operation. The performance comparison of the agent with the baseline controller demonstrated around 21 %, 16 % and 23 % improvement in $PM_{2.5}$ healthy period, thermal comfort period, and energy consumption, respectively, for the DRL agent during the four testing days [31]. Xia et al. [34] proposed a recommender system to co-optimize energy, comfort, and IAQ in commercial buildings. They discussed that the recommender system can potentially reduce energy consumption by up to 8 % in energy-focused optimization and can improve all objectives by 5–10 % in joint optimization compared to rule-based heuristic controllers.

For LE2 (Fig. 3E–H) and HE (Fig. 3I–L), similar trends were observed for the W_1/W_2 ratios under with and without HVAC filter scenarios. However, the absolute exposure reduction and energy penalty values differed due to different emission patterns and ambient conditions (Table 1). Overall, the DOPEEC outperformed DynOpt in terms of energy-efficient exposure reduction for all experimental days and possible scenarios presented in this work, except for the case without the HVAC filter at W_1/W_2 ratio of 1/10, which could again be attributed to the balance of objectives in the reward function and the cost function. The trends in energy penalty and exposure reduction suggest that DOPEEC and similar agents are highly sensitive to the weighted combination in the reward function. The differences between DOPEEC and DynOpt could also be due to the design of the optimization problem. For DynOpt, the optimal solution is directly driven by the cost function and constraints, which is not the case for reinforcement learning algorithms, where some combination of reward function or feedback from the environment governs the optimality, the Q-function in this case. In summary, DOPEEC and similarly trained agents could potentially replace fully physics-driven algorithms such as DynOpt and MPC for real-world adoption since they do not have the same limitations as the latter, and their performance is comparable or even better.

3.3.2. Thermal comfort of occupants

Occupants' thermal comfort is another crucial parameter that must be accounted for to enhance productivity and ensure a safe and comfortable environment inside buildings. Multiple factors, such as indoor temperature, RH, occupancy levels, radiant temperature, and indoor heat emissions, affect the thermal environment inside the buildings, which in turn governs the thermal comfort of the

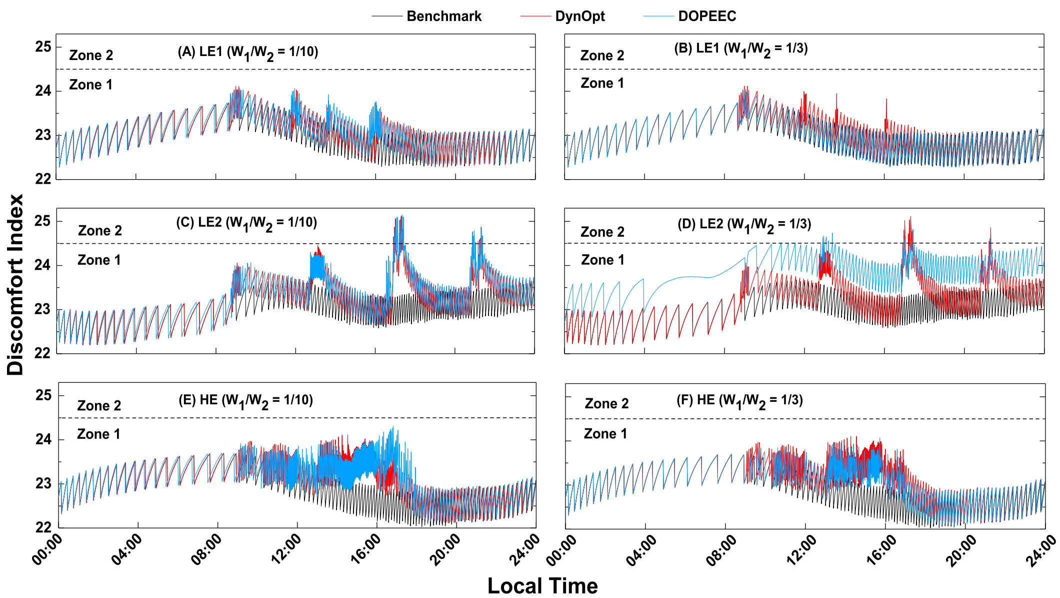

Fig. 4. Comparison of discomfort index (DI) profiles for Agent 1 during different experimental days at two distinct W_1/W_2 ratios.

occupants [53]. Ensuring optimal thermal comfort and exposure reduction with minimal energy penalty is imperative to promote healthy and sustainable buildings [15]. In this work, the thermal model of the house does not account for the number of occupants, metabolic rate, and indoor heat sources and sinks. However, these assumptions are consistent for benchmark and all optimization scenarios (DynOpt and DOPEEC) and thus should not affect the comparative analysis presented here.

Predicted mean vote (PMV) and predicted percentage of dissatisfied (PPD) models are the most used indices for assessing thermal comfort [54–56]. The PPV and PPD indices are governed by factors such as air temperature, air relative humidity, air velocity, mean radiant temperature, activity level, and clothing level [57]. However, the input requirements of PMV-PPD indices are difficult to obtain, and most of the studies have assumed simplified steady-state and uniform conditions for an average person, often due to practical constraints in field measurements, sensor availability, and additional computational cost [58–60]. Prior studies have also utilized simplified models for assessing thermal comfort in indoor environments [34,58,61,62]. Any suitable model for measuring thermal comfort, including the PMV-PPD model, can be incorporated in the proposed control strategy, provided the requisite model inputs are accessible. In this work, a simplified model for discomfort index (DI), a function of indoor temperature and RH, is used to quantify thermal comfort [46,47]. A detailed discussion on the DI estimation and occupants' comfort at different DI levels has been presented in Section S8 of the SI. Briefly, the DI-values have been divided into two zones as per Thom's methodology: (i) Zone 1 ($21<DI<24.5$) indicates less than 50 % of occupants reported discomfort, and Zone 2 ($24.5<DI<27$) means more than 50 % of occupants reported discomfort.

For Agent 1, the indoor set temperature has been kept constant at 25 °C, the same as on experimental days. The DI profiles for benchmark, DynOpt, and DOPEEC without HVAC filter cases are shown in Fig. 4 for all experimental days. Two distinct W_1/W_2 ratios (1/10 and 1/3) have been considered for each day. During all days (LE1, LE2, and HE), the DI remained in Zone 1 at all times for both W_1/W_2 ratios in all scenarios (benchmark, DynOpt, and DOPEEC), except in some instances on LE2, where DI exceeded the Zone 1 limit. The violation of the Zone 1 limit during LE2 could be due to high AERs during high emission activity (Fig. 3E–H), coinciding with high ambient temperature (Fig. S16), which led to increased indoor temperature even with HVAC operating. At $W_1/W_2 = 1/3$, the DI-values of DOPEEC are lower than those of DynOpt, indicating higher thermal comfort. While for LE2, the DI-values are slightly higher than DynOpt, the time duration for Zone1 limit exceedance is less for DOPEEC, which means a lower duration of discomfort to occupants. When the W_1/W_2 ratio is decreased to 1/10, the increase in DI-values for DOPEEC is greater than that of DynOpt. A reverse trend in DI-values is observed compared to the W_1/W_2 ratio of 1/3, as the values for DOPEEC are now slightly higher than DynOpt, and the duration of Zone1 limit exceedance for DOPEEC is longer during LE2. Overall, the indoor environment remained relatively comfortable at all times.

However, as discussed earlier, ensuring this comfort comes at an energy penalty (Table 1). One way to reduce the energy penalty while achieving similar levels of exposure reduction is to increase the indoor set temperature. Mishra et al. [18] have demonstrated that increasing the indoor set temperature by 1 °C does not compromise the thermal comfort of the occupants and reduces the energy penalty significantly. This work introduces an agent (Agent 2) that dynamically changes the indoor set temperature and the indoor-outdoor AER. The performance and operation of Agent 2 are discussed in the next section (Section 3.4). The discussion on the DI profile for HVAC with filter cases (Agent 1) is presented in Section S9 of the SI.

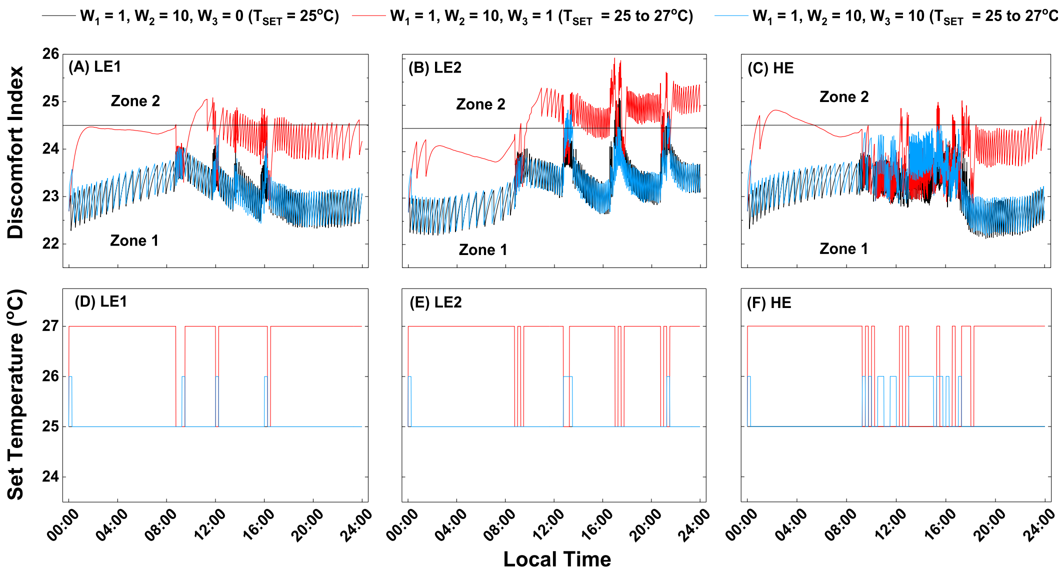

Fig. 5. Comparison of DI profiles (A–C) and dynamic set temperatures (D–F) at different combinations of W_1, W_2, and W_3 for LE1 (A and D), LE2 (B and E), and HE (C and F).

3.4. DOPEEC performance: energy-exposure-thermal comfort optimization by controlling DASS and indoor set temperature (agent 2)

To reduce the energy penalty while achieving similar levels of exposure reduction, Mishra et al. [18] investigated the variation in energy-exposure trade-offs by changing the indoor set temperature from 23 to 28 °C. The study reported that while the exposure reductions remained similar for DynOpt, the energy consumption moved inversely with the set temperature. The previous work considered a constant set temperature throughout the day. However, the energy penalty can be potentially decreased further by dynamic control of the set temperature, which governs HVAC operation. Therefore, this work developed Agent 2, which controls both DASS and set temperature dynamically to minimize indoor pollutant concentration and ensure thermal comfort with minimal energy penalty. For Agent 2, the state space shown in Eq. (5) is modified to include $T_{set,}$ and the action space will be a combination of possible AERs and T_{set} values. In this work, the AER varies from 0.5 ACH to 10 ACH, and T_{set} varies between 25 and 27°C; thus, 60 action combinations are possible (Eq. (6)). Furthermore, a third term is introduced in the reward function for DI, as shown in Eq. (7), to account for dynamic changes in thermal comfort and control action selection. The performance of Agent 2 has been analyzed for two distinct combinations of W_1, W_2, and W_3: (i)1,10,1, and (ii) 1,10,10, and a comparative assessment with Agent 1 has been performed. Fig. 5 demonstrates the DI and set temperature profiles for LE1, LE2, and HE1 when the set temperature is kept at 25 °C (Agent 1) and varies between 25 and 27°C (Agent 2). As discussed earlier, for Agent 1, the DI values remain within Zone 1, while for Agent 2, the interdependency of DI, energy, and exposure, along with their respective weights, determines the AER and set temperature.

For the first combination, i.e., $W_1 = 1$, $W_2 = 10$, and $W_3 = 1$, the DI-values for LE1 and HE remained within Zone 1 most of the time (Fig. 5A and C). In contrast, for LE2 (Fig. 5B), the DI values are well above the threshold after the first emission activity and throughout the remaining day. This observation could be attributed to the coincidence of high emission activities during LE2 with higher outdoor temperatures (Fig. S16), thus leading to a set temperature of 27°C most of the time. For higher AERs corresponding to high PM concentration (Fig. S19), Agent 2 reduces the set temperature to 25 °C (Fig. 5) to maintain some comfort level, though it is insufficient for LE2. The NER values, on the other hand, are considerably greater (LE1: 8.43, LE2: 1.90, HE: 1.07) compared to Agent 1 (LE1: 1.78, LE2: 1.05, HE: 0.79), with slightly lower exposure reductions (Table 2), again demonstrating the trade-off between energy, exposure, and thermal comfort (Table 2). When W_3 is increased to 10, the agent attempts to balance exposure reduction and energy penalty while keeping the DI values within the Zone 1 threshold. Therefore, at $W_3 = 10$, the energy penalty is higher, and exposure reduction is lower for all days compared to the $W_3 = 1$ scenario. The agent's behavior also changes at higher concentrations; the set temperature is 26 °C to reduce the energy penalty compared to 25 °C in the previous scenario (Agent 1).

Fig. S19 shows the indoor PM profile for LE1, LE2, and HE corresponding to Agent 1 at a W_1/W_2 ratio of 1/10 and different combinations of W_1, W_2, and W_3 for Agent 2. The corresponding exposure reduction, energy penalty, and NER values are presented in Table 2. The introduction of DI-term in the reward function (Agent 2) resulted in lower exposure reduction than Agent 1, as shown in Table 2. For LE1, the exposure reduction at $W_3 = 1$ is ~37 % compared to ~33 % at the W_3 value of 10, indicating the lower importance of exposure reduction to reduce the variability in the indoor environment to maintain thermal comfort. The differences in exposure reductions at different W_3-values are more pronounced for the other two experimental days. For LE2, the reductions are ~47 % at $W_3 = 0$, ~42 % at $W_3 = 1$, and 39 % at $W_3 = 10$. For HE, similar exposure reductions are achieved at W_1-values of 0 and 1, while at $W_3 = 10$, an exposure reduction of ~69 % is achieved. On the other hand, in terms of energy penalty, a considerable decline (ranging between ~5 % and ~21 %) is observed for Agent 2 compared to Agent 1 (Table 2). Overall, DOPEEC performed comparable to and even better (in some cases) than DynOpt in optimizing the trade-off between PM exposure and energy consumption while ensuring thermal comfort.

Recent studies have also shown that DRL agents can learn to dynamically vary HVAC air temperature, indoor set temperature, and air velocity when optimizing conflicting objectives [39,41,48,63]. Li et al. [63] designed a DRL agent for demand response-based set temperature control to optimize thermal comfort and electricity cost. The trained agent varied the set temperature as a function of electricity cost, i.e., the set temperature increased to 28 °C to reduce the air conditioning load when the electricity cost was high, and the set temperature lowered as the cost decreased [63]. The set temperature variations observed in the current study also demonstrate a similar pattern, where a high set temperature is maintained to reduce the air conditioning load. Another study by Jian et al. [41] developed a DRL strategy for obtaining indoor set temperatures for multiple thermal zones in real time and demonstrated the agent's capability to vary the indoor set temperatures.

Table 2
Comparison of Exposure, Energy, and NER-value for Agent 2 with different combinations of W_1, W_2, and W_3 at different set temperature variations.

Combinations	Set Temperature Variations	Energy (kWh)			Exposure (μg h/m³)			NER		
		LE1	LE2	HE	LE1	LE2	HE	LE1	LE2	HE
Benchmark	25°C	8.63	10.04	9.19	132.77	176.66	860.76	–	–	–
W1 = 1 W2 = 10 W3 = 0	25°C (Agent 1)	10.56	14.57	17.80	79.75	92.92	224.48	1.78	1.05	0.79
W1 = 1 W2 = 10 W3 = 1	25–27°C (Agent 2)	8.98	12.24	15.53	87.37	102.95	225.13	8.43	1.90	1.07
W1 = 1 W2 = 10 W3 = 10	25–27°C (Agent 2)	10.01	12.45	14.00	88.38	113.34	264.95	2.09	1.49	1.32

4. Limitations and future work

While this study suggests that DOPEEC and similar agents can overcome the challenges associated with purely physics-based strategies such as MPC and DynOpt, more work is required to achieve scalability and optimal performance of DOPEEC. The performance of DRL agents is governed by the robustness of training, which is heavily dependent on the quality and quantity of the dataset [64]. In this work, training of the agent accounting for seasonal variations has not been performed, which could potentially affect the agent's performance under different climatic conditions. Furthermore, the house model for training the agent considers a well-mixed volume; that is not the case in a real-world house, where pollutant concentrations are spatially non-homogeneous [65]. Moreover, a simplified thermal model of the house is utilized to estimate energy consumption and thermal comfort parameters. However, the thermal environment depends on factors such as house characteristics, façade, occupant behavior and choices, and internal heat sources. These assumptions could introduce deviations in the absolute values presented in this work. Therefore, the transferability of RL agents needs to be assessed under varying house characteristics and ambient conditions for their community-scale deployment, which still requires comprehensive testing. Nevertheless, the findings from this study advocate for real-time control of HVAC systems to improve human health and well-being in indoor environments.

5. Conclusion

This work developed a deep reinforcement learning agent for optimizing personal exposure, energy consumption, and thermal comfort inside a house. Two agents (Agent 1 and Agent 2) have been trained, and their performance has been compared to a dynamic optimization (DynOpt) strategy. Since the control via the trained agent could be implemented through low-cost hardware, it has the potential for a large-scale deployment and is not limited by the same constraints as DynOpt and other fully physics-driven control strategies. Agent 1 dynamically controlled AER, while Agent 2 controlled both AER and set temperature to simultaneously reduce pollutant exposure and maintain thermal comfort while minimizing energy penalty. Multiple weighted combinations of objectives in the reward function and HVAC filter scenarios have been designed to evaluate the performance of DOPEEC. The study results demonstrate the capability of these agents to learn the dynamics of the indoor environment under different emission activities and ambient conditions. Similar to DynOpt, the performance and control parameters in DOPEEC highly depend on the relative importance of individual objectives in the reward function. While DynOpt is less sensitive to small changes in the relative importance of objectives in the cost function, the performance of DOPEEC changes drastically for the same amount of change in the reward function.

For Agent 1, the NER values are ~26 %–~83 % higher than DynOpt for different emission activities with an HVAC filter installed at both W_1/W_2 ratios (1/3 and 1/10). For each emission activity, at the W_1/W_2 ratio of 1/10 in the absence of an HVAC filter, the NER values for Agent 1 are ~17 %–39 % lower than DynOpt. The lower NER values for Agent 1 are observed due to an average 10 % higher energy penalty that accompanied a slightly higher average exposure reduction (~7 %) than DynOpt. Furthermore, the DI values for Agent 1 remained within Zone 1 (21–24.5) at all times except minor violations at around 17:00 when elevated indoor PM levels coincided with high ambient temperature. When the set temperature is included in the control actions (Agent 2), NER improved further with ~17 %–~373 % increase compared to Agent 1, while the PM exposure remained similar. Moreover, an increased importance of thermal comfort in the reward function ($W_3 = 10$) led to decreased Zone 1 violations of DI during different emission activities.

The performance of DRL agents presented in this work demonstrates their potential for a solution that can be deployed at a large scale to improve IAQ while ensuring thermal comfort with optimized energy penalty. The trained agent could be uploaded on a microcontroller, and real-time data collected using low-cost monitors could be fed to the agent to implement real-time control actions for ventilation and air conditioning units. Simultaneously, the ability to dynamically adjust the set temperature based on the thermal comfort levels is consequential for occupant-centric control, accounting for behavioral preferences, occupancy levels, and personal choices.

CRediT authorship contribution statement

Nishchaya Kumar Mishra: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Nipun Batra:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Sameer Patel:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jobe.2025.114074.

Data availability

Data will be made available on request.

References

[1] A.H. Goldstein, W.W. Nazaroff, C.J. Weschler, J. Williams, How do indoor environments affect air pollution exposure? Environ. Sci. Technol. 55 (2021) 100–108, https://doi.org/10.1021/ACS.EST.0C05727.
[2] J.G. Allen, J.D. Macomber, Healthy Buildings: How Indoor Space can Make you Sick or Keep you Well, 2022.
[3] C.H. Jeong, S. Salehi, J. Wu, M.L. North, J.S. Kim, C.W. Chow, et al., Indoor measurements of air pollutants in residential houses in urban and suburban areas: indoor versus ambient concentrations, Sci. Total Environ. 693 (2019) 133446, https://doi.org/10.1016/J.SCITOTENV.2019.07.252.
[4] T.V. Vu, G.B. Stewart, N. Kitwiroon, S. Lim, B. Barratt, F.J. Kelly, et al., Assessing the contributions of outdoor and indoor sources to air quality in London homes of the SCAMP cohort, Build. Environ. 222 (2022) 109359, https://doi.org/10.1016/J.BUILDENV.2022.109359.
[5] P. Azimi, B. Stephens, A framework for estimating the US mortality burden of fine particulate matter exposure attributable to indoor and outdoor microenvironments, J. Expo. Sci. Environ. Epidemiol. 30 (2) (2018) 271–284, https://doi.org/10.1038/s41370-018-0103-4, 2018 30.
[6] D. Majra, J. Benson, J. Pitts, J. Stebbing, SARS-CoV-2 (COVID-19) superspreader events, J. Infect. 82 (2021) 36–40, https://doi.org/10.1016/J.JINF.2020.11.021.
[7] S. Rayegan, C. Shu, J. Berquist, J. Jeon, L. Zhou, L. Wang, et al., A review on indoor airborne transmission of COVID-19– modelling and mitigation approaches, J. Build. Eng. 64 (2023) 105599, https://doi.org/10.1016/J.JOBE.2022.105599.
[8] L. Morawska, J.W. Tang, W. Bahnfleth, P.M. Bluyssen, A. Boerstra, G. Buonanno, et al., How can airborne transmission of COVID-19 indoors be minimised? Environ. Int. 142 (2020) 105832 https://doi.org/10.1016/J.ENVINT.2020.105832.
[9] L. Morawska, Y. Li, T. Salthammer, Lessons from the COVID-19 pandemic for ventilation and indoor air quality, Science 385 (2024) 396–401, https://doi.org/10.1126/SCIENCE.ADP2241, 1979.
[10] T. Pistochini, C. Mande, S. Chakraborty, Modeling impacts of ventilation and filtration methods on energy use and airborne disease transmission in classrooms, J. Build. Eng. 57 (2022) 104840, https://doi.org/10.1016/J.JOBE.2022.104840.
[11] L.A. Wallace, T. Zhao, N.E. Klepeis, Indoor contribution to PM2.5 exposure using all PurpleAir sites in Washington, Oregon, and California, Indoor Air 32 (2022) e13105, https://doi.org/10.1111/INA.13105.
[12] D.M. Lunderberg, Y. Liang, B.C. Singer, J.S. Apte, W.W. Nazaroff, A.H. Goldstein, Assessing residential PM2.5 concentrations and infiltration factors with high spatiotemporal resolution using crowdsourced sensors, Proc. Natl. Acad. Sci. U. S. A. 120 (2023) e2308832120, https://doi.org/10.1073/PNAS.2308832120.
[13] X. Cao, X. Dai, J. Liu, Building energy-consumption status worldwide and the state-of-the-art technologies for zero-energy buildings during the past decade, Energy Build. 128 (2016) 198–213, https://doi.org/10.1016/J.ENBUILD.2016.06.089.
[14] M. Santamouris, K. Vasilakopoulou, Present and future energy consumption of buildings: challenges and opportunities towards decarbonisation, E-Prime - Advances in Electrical Engineering, Electronics and Energy 1 (2021) 100002, https://doi.org/10.1016/J.PRIME.2021.100002.
[15] N.K. Mishra, S. Patel, Need for a holistic approach to assessing sustainable, green, and healthy buildings, Environ. Health (2024), https://doi.org/10.1021/ENVHEALTH.4C00161.
[16] B. Pandey, B. Bohara, R. Pungaliya, S.C. Patwardhan, R. Banerjee, A thermal comfort-driven model predictive controller for residential split air conditioner, J. Build. Eng. 42 (2021) 102513, https://doi.org/10.1016/J.JOBE.2021.102513.
[17] H.S. Ganesh, K. Seo, H.E. Fritz, T.F. Edgar, A. Novoselac, M. Baldea, Indoor air quality and energy management in buildings using combined moving horizon estimation and model predictive control, J. Build. Eng. 33 (2021), https://doi.org/10.1016/j.jobe.2020.101552.
[18] N.K. Mishra, M.E. Vance, A. Novoselac, S. Patel, Dynamic optimization of personal exposure and energy consumption while ensuring thermal comfort in a test house, Build. Environ. 252 (2024) 111265, https://doi.org/10.1016/J.BUILDENV.2024.111265.
[19] S. Zhang, Z. Ai, Z. Lin, Novel demand-controlled optimization of constant-air-volume mechanical ventilation for indoor air quality, durability and energy saving, Appl. Energy 293 (2021) 116954, https://doi.org/10.1016/J.APENERGY.2021.116954.
[20] H.S. Ganesh, H.E. Fritz, T.F. Edgar, A. Novoselac, M. Baldea, A model-based dynamic optimization strategy for control of indoor air pollutants, Energy Build. 195 (2019) 168–179, https://doi.org/10.1016/j.enbuild.2019.04.022.
[21] X. Sha, Z. Ma, S. Sethuvenkatraman, W. Li, Online learning-enhanced data-driven model predictive control for optimizing HVAC energy consumption, indoor air quality and thermal comfort, Appl. Energy 383 (2025) 125341, https://doi.org/10.1016/J.APENERGY.2025.125341.
[22] J. Wang, S. Li, H. Chen, Y. Yuan, Y. Huang, Data-driven model predictive control for building climate control: three case studies on different buildings, Build. Environ. 160 (2019) 106204, https://doi.org/10.1016/J.BUILDENV.2019.106204.
[23] D. Zhuang, V.J.L. Gan, Z. Duygu Tekler, A. Chong, S. Tian, X. Shi, Data-driven predictive control for smart HVAC system in IoT-integrated buildings with time-series forecasting and reinforcement learning, Appl. Energy 338 (2023) 120936, https://doi.org/10.1016/J.APENERGY.2023.120936.
[24] P. Stoffel, M. Berktold, D. Müller, Real-life data-driven model predictive control for building energy systems comparing different machine learning models, Energy Build. 305 (2024) 113895, https://doi.org/10.1016/J.ENBUILD.2024.113895.
[25] S. Miao, M. Gangolells, B. Tejedor, Data-driven model for predicting indoor air quality and thermal comfort levels in naturally ventilated educational buildings using easily accessible data for schools, J. Build. Eng. 80 (2023) 108001, https://doi.org/10.1016/J.JOBE.2023.108001.
[26] H. Lee, Y. Heo, Simplified data-driven models for model predictive control of residential buildings, Energy Build. 265 (2022) 112067, https://doi.org/10.1016/J.ENBUILD.2022.112067.
[27] T. Yang, L. Zhao, W. Li, J. Wu, A.Y. Zomaya, Towards healthy and cost-effective indoor environment management in smart homes: a deep reinforcement learning approach, Appl. Energy 300 (2021) 117335, https://doi.org/10.1016/J.APENERGY.2021.117335.
[28] F. Guo, S woo Ham, D. Kim, H.J. Moon, Deep reinforcement learning control for co-optimizing energy consumption, thermal comfort, and indoor air quality in an office building, Appl. Energy 377 (2025) 124467, https://doi.org/10.1016/J.APENERGY.2024.124467.
[29] W. Shang, J. Liu, C. Wang, J. Li, X. Dai, Developing smart air purifier control strategies for better IAQ and energy efficiency using reinforcement learning, Build. Environ. 242 (2023) 110556, https://doi.org/10.1016/J.BUILDENV.2023.110556.
[30] W. Valladares, M. Galindo, J. Gutiérrez, W.C. Wu, K.K. Liao, J.C. Liao, et al., Energy optimization associated with thermal comfort and indoor air control via a deep reinforcement learning algorithm, Build. Environ. 155 (2019) 105–117, https://doi.org/10.1016/J.BUILDENV.2019.03.038.
[31] Y. An, C. Chen, Energy-efficient control of indoor PM2.5 and thermal comfort in a real room using deep reinforcement learning, Energy Build. 295 (2023) 113340, https://doi.org/10.1016/J.ENBUILD.2023.113340.
[32] Y. An, T. Xia, R. You, D. Lai, J. Liu, C. Chen, A reinforcement learning approach for control of window behavior to reduce indoor PM2.5 concentrations in naturally ventilated buildings, Build. Environ. 200 (2021) 107978, https://doi.org/10.1016/J.BUILDENV.2021.107978.

[33] S.K. Heo, K.J. Nam, J. Loy-Benitez, Q. Li, S.C. Lee, C.K. Yoo, A deep reinforcement learning-based autonomous ventilation control system for smart indoor air quality management in a subway station, Energy Build. 202 (2019) 109440, https://doi.org/10.1016/J.ENBUILD.2019.109440.
[34] S. Xia, P. Wei, Y. Liu, A. Sonta, X. Jiang, RECA: a multi-task deep reinforcement learning-based recommender system for Co-Optimizing energy, comfort and air quality in commercial buildings, in: BuildSys 2023 - Proceedings of the10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation, Association for Computing Machinery, Inc, 2023, pp. 99–109, https://doi.org/10.1145/3600100.3623735.
[35] J.R. Vázquez-Canteli, Z. Nagy, Reinforcement learning for demand response: a review of algorithms and modeling techniques, Appl. Energy 235 (2019) 1072–1089, https://doi.org/10.1016/J.APENERGY.2018.11.002.
[36] M. Han, R. May, X. Zhang, X. Wang, S. Pan, Y. Da, et al., A novel reinforcement learning method for improving occupant comfort via window opening and closing, Sustain. Cities Soc. 61 (2020) 102247, https://doi.org/10.1016/J.SCS.2020.102247.
[37] A.K. Shakya, G. Pillai, S. Chakrabarty, Reinforcement learning algorithms: a brief survey, Expert Syst. Appl. 231 (2023) 120495, https://doi.org/10.1016/J.ESWA.2023.120495.
[38] Q. Chen, Q. Zhang, Y. Liu, Balancing exploration and exploitation in episodic reinforcement learning, Expert Syst. Appl. 231 (2023) 120801, https://doi.org/10.1016/J.ESWA.2023.120801.
[39] L. Yao, L.Y. Huang, J.C. Teo, HVAC control based on reinforcement learning and fuzzy reasoning: optimizing HVAC supply air temperature, flow rate, and velocity, J. Build. Eng. 103 (2025) 112143, https://doi.org/10.1016/J.JOBE.2025.112143.
[40] K. Al Sayed, A. Boodi, R. Sadeghian Broujeny, K. Beddiar, Reinforcement learning for HVAC control in intelligent buildings: a technical and conceptual review, J. Build. Eng. 95 (2024) 110085, https://doi.org/10.1016/J.JOBE.2024.110085.
[41] B. Jiang, J. Wang, Y. Li, P. Wang, Y. Rezgui, C. Zhang, et al., Dynamic optimization of cooling temperature setpoints in multiple thermal zones of office buildings in connection with D3QN, Energy Build. 341 (2025) 115851, https://doi.org/10.1016/J.ENBUILD.2025.115851.
[42] D.K. Farmer, M.E. Vance, J.P.D. Abbatt, A. Abeleira, M.R. Alves, C. Arata, et al., Overview of HOMEChem: House observations of microbial and environmental chemistry, Environ Sci Process Impacts 21 (2019) 1280–1300, https://doi.org/10.1039/C9EM00228F.
[43] G. Brockman, V. Cheung, L. Pettersson, J. Schneider, J. Schulman, J. Tang, et al., OpenAI gym. https://doi.org/10.48550/arXiv.1606.01540, 2016.
[44] S. Patel, D. Rim, S. Sankhyan, A. Novoselac, M.E. Vance, Aerosol dynamics modeling of sub-500 nm particles during the HOMEChem study, Environ Sci Process Impacts 23 (2021) 1706–1717, https://doi.org/10.1039/D1EM00259G.
[45] S. Patel, S. Sankhyan, E.K. Boedicker, P.F. Decarlo, D.K. Farmer, A.H. Goldstein, et al., Indoor particulate matter during HOMEChem: concentrations, size distributions, and exposures, Environ. Sci. Technol. 54 (2020) 7107–7116, https://doi.org/10.1021/ACS.EST.0C00740.
[46] T.A. Yousif, H.M.H. Tahir, Application of thom's thermal discomfort index in Khartoum state, Sudan, J. For. Prod. Ind. (2013). https://www.researchgate.net/publication/272623320_Application_of_Thom's_Thermal_Discomfort_Index_in_Khartoum_State_Sudan. (Accessed 7 July 2023).
[47] E.C. Thom, The Discomfort Index, vol. 12, 2010, pp. 57–61, https://doi.org/10.1080/00431672.1959.9926960. Https://DoiOrg/101080/0043167219599926960.
[48] A. Azimi, O. Akbari, A deep reinforcement learning-based method for dynamic quality of service aware energy and occupant comfort management in intelligent buildings, E-Prime - Advances in Electrical Engineering, Electronics and Energy 9 (2024) 100700, https://doi.org/10.1016/J.PRIME.2024.100700.
[49] M. Liu, M. Guo, Y. Fu, Z. O'Neill, Y. Gao, Expert-guided imitation learning for energy management: evaluating GAIL's performance in building control applications, Appl. Energy 372 (2024) 123753, https://doi.org/10.1016/J.APENERGY.2024.123753.
[50] S. Dey, T. Marzullo, X. Zhang, G. Henze, Reinforcement learning building control approach harnessing imitation learning, Energy and AI 14 (2023) 100255, https://doi.org/10.1016/J.EGYAI.2023.100255.
[51] S. Liu, X. Liu, T. Zhang, C. Wang, W. Liu, Joint optimization for temperature and humidity independent control system based on multi-agent reinforcement learning with cooperative mechanisms, Appl. Energy 375 (2024) 123968, https://doi.org/10.1016/J.APENERGY.2024.123968.
[52] A.J. Khabbazi, E.N. Pergantis, L.D. Reyes Premer, P. Papageorgiou, A.H. Lee, J.E. Braun, et al., Lessons learned from field demonstrations of model predictive control and reinforcement learning for residential and commercial HVAC: a review, Appl. Energy 399 (2025) 126459, https://doi.org/10.1016/J.APENERGY.2025.126459.
[53] K. Karyono, B.M. Abdullah, A.J. Cotgrave, A. Bras, The adaptive thermal comfort review from the 1920s, the present, and the future, Developments in the Built Environment 4 (2020) 100032, https://doi.org/10.1016/J.DIBE.2020.100032.
[54] M. Elhami, S.S. Goodarzi, S. Maleki, B. Sajadi, Three-objective optimization of the HVAC system control strategy in an educational building to reduce energy consumption and enhance indoor environmental quality (IEQ) using machine learning techniques, J. Build. Eng. 105 (2025) 112444, https://doi.org/10.1016/J.JOBE.2025.112444.
[55] T. Cheung, S. Schiavon, T. Parkinson, P. Li, G. Brager, Analysis of the accuracy on PMV – PPD model using the ASHRAE global thermal comfort database II, Build. Environ. 153 (2019) 205–217, https://doi.org/10.1016/J.BUILDENV.2019.01.055.
[56] E.E. Broday, C.R. Ruivo, M. Gameiro da Silva, The use of Monte Carlo method to assess the uncertainty of thermal comfort indices PMV and PPD: benefits of using a measuring set with an operative temperature probe, J. Build. Eng. 35 (2021) 101961, https://doi.org/10.1016/J.JOBE.2020.101961.
[57] Standard 55 – thermal environmental conditions for human occupancy n.d. https://www.ashrae.org/technical-resources/bookstore/standard-55-thermal-environmental-conditions-for-human-occupancy. (Accessed 6 July 2023).
[58] H. Lee, Y. Heo, Simplified data-driven models for model predictive control of residential buildings, Energy Build. 265 (2022) 112067, https://doi.org/10.1016/J.ENBUILD.2022.112067.
[59] B. Pandey, B. Bohara, R. Pungaliya, S.C. Patwardhan, R. Banerjee, A thermal comfort-driven model predictive controller for residential split air conditioner, J. Build. Eng. 42 (2021) 102513, https://doi.org/10.1016/J.JOBE.2021.102513.
[60] T. Cheung, S. Schiavon, T. Parkinson, P. Li, G. Brager, Analysis of the accuracy on PMV – PPD model using the ASHRAE global thermal comfort database II, Build. Environ. 153 (2019) 205–217, https://doi.org/10.1016/J.BUILDENV.2019.01.055.
[61] N. Aqilah, H.B. Rijal, S.A. Zaki, Indoor thermal environment and effect of air movement on comfort temperature in Malaysian naturally ventilated dwellings, J. Build. Eng. 104 (2025) 112151, https://doi.org/10.1016/J.JOBE.2025.112151.
[62] S. Miao, M. Gangolells, B. Tejedor, Data-driven model for predicting indoor air quality and thermal comfort levels in naturally ventilated educational buildings using easily accessible data for schools, J. Build. Eng. 80 (2023) 108001, https://doi.org/10.1016/J.JOBE.2023.108001.
[63] Z. Li, Z. Sun, Q. Meng, Y. Wang, Y. Li, Reinforcement learning of room temperature set-point of thermal storage air-conditioning system with demand response, Energy Build. 259 (2022) 111903, https://doi.org/10.1016/J.ENBUILD.2022.111903.
[64] G. Dulac-Arnold, N. Levine, D.J. Mankowitz, J. Li, C. Paduraru, S. Gowal, et al., Challenges of real-world reinforcement learning: definitions, benchmarks and analysis, Mach. Learn. 110 (2021) 2419–2468, https://doi.org/10.1007/S10994-021-05961-4.
[65] A.K. Thakur, S. Patel, Characterization of particulate matter in a multizonal residential apartment: transport, exposure, and mitigation, Environ. Sci. J. Integr. Environ. Res.: Atmosphere 4 (2024) 1026–1041, https://doi.org/10.1039/D4EA00080C.